

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1047

Gardner 230 kV
October 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1047. The Interconnection Customer proposes a 743.2 MW natural gas combined cycle facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on August 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$70,536,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wabash 500-230 kV Autotransformer	Dekalb – Richland 500.0 kV	163.1	1,022	19.5
Top Crop 500/230kV Transformer	Grand Ridge Battery Projects – Gardner 230.0 kV	143.8	1,350	95.0
Gardner – Grand Ridge Battery Projects 230.0 kV	Top Crop – Gardner 230.0 kV line	139.6	1,521	99.9
Top Crop – Gardner 230.0 kV Line	Grand Ridge Battery Projects – Gardner 230.0 kV	139.6	1,521	99.9
Treat – Camberlin 230.0 kV	El Paso – Ravenna 500.0 kV Line	135.7	633	5.8
Montpelier Electric Generating Sta 500/230kV Autotransformer	Richland – Dekalb 500.0 kV	134.9	1,479	14.1
Hebo – Barrett 230.0 kV Line	Winthrop – Hebo 230.0 kV Line	130.9	873	5.7
Baraga Vlg 500/230 kV TX	Beaver – Trego 500.0 kV	130.3	1,689	13.2
Myrtle Pt – George Neal S 230.0 kV	Trego – Galion Generating Station 500.0 kV line	128.8	974	6.5
Zona Libre 6007 – George Neal South 230.0 kV	Trego Substation – Galion Generating Station 500.0 kV	127.9	1,001	6.5
Stoner – Ledges 230.0 kV L	Ravenna – El Paso 500.0 kV	127.9	618	5.8
Madeline – E Methuen 230.0 kV	Gray'S Flat – E Methuen 230.0 kV	127.5	1,495	9.5
Waldo 500/230kV XFMR	Trego – Beaver 500.0 kV	127.4	1,248	9.8
Kampsville – Zona Libre 6007 230.0 kV line	Trego – Galion Generating Sta 500.0 kV	127.1	1,028	6.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chamberlin – Involta 230.0 kV Line	Hylebos – Ravenna 500.0 kV L	126.9	500	5.3
Madeline – Baraga Vlg 230.0 kV	E Methuen – Gray'S Flat 230.0 kV	126.7	963	6.5
Kampsville – Moreno Tap 230.0 kV	Galion Generating Sta – Trego 500.0 kV Line	126.6	1,044	6.5
Hillcrest – Segs 5 100.0 kV	Richland – Dekalb 500.0 kV	124.4	554	7.2
Trego 230/1kV Auto	Troy Energy Llc – Trego 500.0 kV	123.4	829	13.0
James De Young – Burris Pit Substation 230.0 kV	Trego – Beaver 500.0 kV	123.3	1,392	7.9
Dekalb Substation – Robison Park 230.0 kV line	Dekalb – Jones Chapel 500.0 kV	123.1	2,339	34.1
Darlington Road – Wayne Trace 100.0 kV	Richland – Dekalb 500.0 kV line	123.1	461	6.0
Atascocita – North Cambridge 230.0 kV	El Paso – Ravenna Substation 500.0 kV	122.9	1,216	9.1
E Methuen – Lonetree 500.0 kV	Montpelier Electric Generating Sta – Muskogee 500.0 kV	122.1	3,951	9.2
Sabanera – Lonetree 500.0 kV	Muskogee – Montpelier Electric Generating Sta 500.0 kV	121.9	4,315	10.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wabash 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$55,029,000	6% (\$3,480,000)
Top Crop 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$24,121,000	39% (\$9,371,000)
Gardner – Grand Ridge Battery Projects 230.0 kV	Rebuild line with higher-capacity conductor	\$13,533,000	39% (\$5,258,000)
Top Crop – Gardner 230.0 kV Line	Rebuild line with higher-capacity conductor	\$19,695,000	39% (\$7,651,000)
Chamberlin – Treat 230.0 kV	Rebuild line with higher-capacity conductor	\$27,049,000	6% (\$1,706,000)
Montpelier Electric Generating Sta Substation 500/230 kV Auto	Replace transformer with higher-rated unit	\$26,014,000	6% (\$1,645,000)
Hebo – Barrett Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$8,650,000	6% (\$547,000)
Baraga Vlg 500/230 kV TX	Replace transformer with higher-rated unit	\$33,226,000	6% (\$2,102,000)
MyrtlePt – George Neal S 230.0 kV	Rebuild line with higher-capacity conductor	\$22,022,000	6% (\$1,394,000)
George Neal S – Zona Libre 6007 230.0 kV	Rebuild line with higher-capacity conductor	\$62,779,000	6% (\$3,975,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stoner – Ledges 230.0 kV L	Rebuild line with higher-capacity conductor	\$30,969,000	6% (\$1,953,000)
Madeline – E Methuen 230.0 kV	Rebuild line with higher-capacity conductor	\$17,540,000	6% (\$1,111,000)
Waldo 500/230kV XFMR	Replace transformer with higher-rated unit	\$36,041,000	6% (\$2,284,000)
Zona Libre 6007 – Kampsville 230.0 kV L	Rebuild line with higher-capacity conductor	\$23,459,000	6% (\$1,485,000)
Chamberlin – Involta 230.0 kV	Rebuild line with higher-capacity conductor	\$15,898,000	6% (\$1,005,000)
Madeline – Baraga Vlg 230.0 kV	Rebuild line with higher-capacity conductor	\$17,849,000	6% (\$1,130,000)
Moreno Tap Substation – Kampsville 230.0 kV	Rebuild line with higher-capacity conductor	\$42,089,000	6% (\$2,665,000)
Hillcrest – Segs 5 100.0 kV	Rebuild line with higher-capacity conductor	\$2,685,000	6% (\$170,000)
Trego 230/1kV Auto	Replace transformer with higher-rated unit	\$8,426,000	6% (\$535,000)
James De Young – Burris Pit Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$58,175,000	6% (\$3,681,000)
Dekalb Substation – Robison Park 230.0 kV line	Rebuild line with higher-capacity conductor	\$38,412,000	6% (\$2,428,000)
Wayne Trace – Darlington Rd 100.0 kV	Rebuild line with higher-capacity conductor	\$1,518,000	6% (\$96,000)
North Cambridge Substation – Atascocita 230.0 kV	Rebuild line with higher-capacity conductor	\$21,530,000	6% (\$1,358,000)
E Methuen – Lonetree 500.0 kV line	Rebuild line with higher-capacity conductor	\$99,957,000	6% (\$6,309,000)
Lonetre – Sabanera 500.0 kV	Rebuild line with higher-capacity conductor	\$114,107,000	6% (\$7,197,000)
	Total allocated to project		\$70,536,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.